

Saliency difference based objective evaluation method for a superimposed screen of the HUD with various background

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Abstract: The head-up display (HUD) is an emerging device that can project information on a transparent screen. This device is used in airplanes and vehicles and is usually placed in the operator's field of view. In the case of a vehicle, the driver can visualize not only a wide variety of information on the HUD, but also the background (driving environment) through the HUD. However, the projected information on the display may interfere with the colors in the background because the device is transparent. For example, a text message in red color on the HUD has a lower contrast when there the background is also red, such as the red brake lights of a vehicle. As a first step in solving this issue, it is necessary to evaluate the mutual interference between the information on the HUD and the background. In this regard, a method to evaluate mutual interference based on saliency is proposed. This information can be evaluated by comparing the HUD component in a saliency map of a measured image with the HUD image.

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Keywords: Human-machine interface, Efficient evaluation, Graphic displays

1. INTRODUCTION

The amount of information that an in-vehicle information system (IVIS) can provide to a driver has increased rapidly with the development of information technologies for vehicles. A head-up display (HUD) is a transparent display that projects information from the IVISs such as advanced driver assistance systems (ADASs) (Thompson et al. (2006)) and car-navigation systems (Kazimierski et al. (2017)) to the drivers field of view. This device allows the driver to simultaneously visualize both the information on the HUD and the background because the HUD is transparent. The reaction time of the driver is therefore improved compared to the case when the information is presented on a multi-information display on a meter cluster because the distance and the time of the eye movement are reduced. Consequently, the HUD is widely used in the field of aviation, such as in civilian aircraft and fighter jets (Arthur et al. (2005)).

However, HUDs also have obvious disadvantages because of their transparency. The driver may experience difficulty in visualizing projected information such as texts, icons, and graphs when it overlaps with a background of the same color. Most of HUDs used in airplanes are monochromatic, e.g., green or orange (Wood and Howells (2000)), because there is typically a limited range of colors in the sky

background (white for the clouds, blue for the sky, black for the night sky). In the case of a vehicle, there is a wide variety of objects with different colors and shapes such as buildings, trees, other vehicles, pedestrians, and even traffic markers on the road. The driver must visually recognize different information and objects in a traffic environment. The environment on the ground is more complex than that in the sky, and therefore, the potential overlap of the background with the display information might affect the driver's ability to recognize visual elements in the HUD. However, in the case when the projected information is too prominent, it may reduce the visibility of objects in the background that required the attention of the driver. This may result in a potentially dangerous situation, especially in an emergent case such as rush-out of a pedestrian in front of a vehicle.

Many studies have analyzed the effect of HUDs on the driver from different perspectives such as eye movement (Weintraub et al. (1985)), focusing ability (Charissis and Naef (2007)), information visibility (Inuzuka et al. (1991)), and response times (Enomoto et al. (2016); Wolffsohn et al. (1998)). The evaluation of the subject was utilized in most studies. Participant feedback is directly obtained but this requires excessive time and manpower. By contrast, objective evaluations without subject exper-

iments could reduce both manpower and time. The objective evaluation method generally evaluates objects based on a hypothesis that simulates the participant's feedback.

In this paper, a method is proposed to evaluate the mutual effect between projected information on the HUD and the background. This approach is based on an individual's visual perception process. Specifically, this paper proposes a very simple objective evaluation method using the concept of visual saliency. Visual saliency is an early vision mechanism for identifying objects that attract attention (Itti and Koch (2000)). There is a high probability that the driver notices projected information on the HUD when it is high. The contributions of this study are as follows:

- the mutual effect between the background and the projected information on the HUD is objectively analyzed.
- the proposed method would help HUD designers to quickly evaluate and intuitively determine whether the human-machine interface (HMI) is easily recognized by users under certain circumstance.

2. BACKGROUND

2.1 Evaluations for HUD

The HUD has been investigated in many studies from different perspectives with respect to its effect on drivers. Inuzuka et al. (1991) evaluated the effect of HUDs on users from a cognitive information perspective. It has been considered that several factors affect the cognition of information on these devices such as location, character size, brightness, and color. In addition, the gaze point distribution, recognition time, and subjective evaluation were investigated when the participants used a HUD. The results showed that the recognition time when a HUD is used is significantly shorter compared to the use of a conventional instrument panel display. This study examines the advantages of HUDs, however, several studies have noted that the effectiveness of these devices may be influenced by the background information when drivers are in complex environments.

Enomoto et al. (2016) investigated the reaction time of a driver's braking behavior during HUD usage in the case when a brake light of a preceding vehicle is turned on. A driving simulator was used to perform a subject experiment. The results showed that the braking reaction time of the driver was reduced when the projected information on the HUD overlapped with the brake light of the preceding vehicle. Charissis and Naef (2007) used a full-windshield display to analyze the focusing ability of a driver based on subject experiments. The Participants encountered difficulties in focusing when the projection distance was reduced. Most of the participants relied on the HUD for driving in low visibility scenarios such as snow. In this case, the colors were undiversified and the lines became indistinct in the background. Moreover, the projected information on the HUD becomes more salient. They also considered that the design of the HUD must minimize visual clutter and distraction under high visibility conditions. Wolffsohn et al. (1998) analyzed the response times and the detection rate when both the HUD image and the background scene changed. They determined that the HUD could reduce the eye movement distance of drivers, but the cognitive

demand required by the HUD task may distract the driver from observing the outside world and cause the drivers' response times to be longer. This problem will become more pronounced with age.

2.2 Saliency map methods

Visual saliency map methods can be generally divided into two types: a top-down type and a bottom-up type. If an individual has no prior predictions about the appearance of an object or cannot actively find it, then this is an example of bottom-up saliency. Conversely, if an individual takes the initiative to search for an object or has prior knowledge of the appearance of the object, then this represents top-down saliency.

For bottom-up saliency, Itti et al. (1998) proposed a basic method. It simulates the early vision process of an individual when his/her attention is diverted to an object (Itti and Koch (2000)). Specifically, the early vision process is initiated when the light hits the retina and the photoreceptor cells are activated until the ganglion cells are activated. Itti et al. (1998)'s method considers three elements of visual saliency including intensity, color, and orientation. They mimicked contrast detection in the receptive field and proposed center-surround differences and an across-scale summation as part of their method. On the basis of previous studies, Itti and Koch (2000) considered the influence of dynamic changes of the image on visual saliency and added elements of motion in saliency map methods. Walther and Koch (2006) proposed a bottom-up saliency method, which is a biologically plausible model to form and address to proto-objects in natural scenes. This model can be used to serialize the visual processing by the biologically plausible model of object recognition to address proto-objects. Achanta et al. (2009) proposed a bottom-up saliency method that is a frequency tuned approach for computing saliency in images using low-level features of color and luminance.

In the case of top-down saliency, many models used machine learning approaches to train the parameters of the saliency models. When training the model, the position of the object in the image is often used as the label, or the gaze information of human is directly used as the label. Kavak et al. (2013) considered saliency estimation as a supervised learning problem and accordingly, applied machine learning. They proposed a top-down saliency model based on multiple bottom-up saliency methods with fixation maps and used a multiple kernel learning method to train this model. Deep learning methods were also used to train the top-down saliency model. Pan and Giró-i Nieto (2015) proposed an end-to-end saliency model based on a convolutional network. They used fixation information as the label, similar to Kavak et al. (2013). This model had a good performance for saliency prediction because the convolutional network had a strong ability to extract local features of images.

In all of these methods, it is difficult to evaluate the mutual interference of saliency between the HUD and background, although the saliency of an image is readily obtained. Therefore, this paper proposes a method to address mutual interference, that can be applied to any top-down and the bottom-up saliency method.

3. PROPOSED METHOD

This paper proposes two indices to evaluate the mutual interference of saliency between the HUD and background. The first index is the degree of reduction in the saliency of the projected information due to the effect of the background on the HUD. The second one is the degree of visual distraction due to the effect of the background on the HUD.

The general process of the proposed method is shown in Fig. 1. Firstly, when using the HUD, the driver's entire field of view is called a measured image $\mathbf{M} \in \mathbb{Z}^{+W \times H \times C}$, where W and H are the width and height of this image, and C is the color channel of the image. Then, a saliency map $\mathbf{M}^S \in \mathbb{Z}^{+W \times H}$ is computed from $\mathbf{M}$.

Subsequently, the HUD component is extracted from the $\mathbf{M}^S$ because this part is the target to be evaluated. It is described as $\mathbf{M}^{S,HUD} \in \mathbb{Z}^{N \times M}$, where N and M are the height and width of $\mathbf{M}^{S,HUD}$. The input image of the HUD is called a HUD image $\mathbf{H} \in \mathbb{Z}^{N \times M \times C}$. In this paper, the background of the HUD image is set to be black because of the following two reasons. Firstly, the black color that is projected on the HUD will become transparent, and the transparency is one of the advantages of the HUD. Secondly, this study considers that the saliency of colored information on a black background is relatively high.

The saliency map of the HUD image is then generated and consistently resized using the size of $\mathbf{M}^{S,HUD}$ and is denoted as $\mathbf{H}^S \in \mathbb{Z}^{N \times M}$. It will be used as a baseline to evaluate the mutual effect between the HUD and the background. Therefore, the difference $\mathbf{E}$ between $\mathbf{M}^{S,HUD}$ and $\mathbf{H}^S$ is calculated as

$$\mathbf{E} = \mathbf{M}^{S,HUD} - \mathbf{H}^S \in \mathbb{Z}^{N \times M}. \quad (1)$$

The plus element component of $\mathbf{E}$ is written as $\mathbf{E}^{plus}$, and the minus component is written as $\mathbf{E}^{minus}$. Therefore, each element of these two components is computed as

$$E_{i,j}^{plus} = \max(E_{i,j}, 0) \quad (2)$$

$$E_{i,j}^{minus} = -\min(E_{i,j}, 0) \quad (3)$$

It should be noted that $\mathbf{E}^{plus}$ represents the saliency in the display field of the HUD except for the projected information, and it is the visual distraction derived from the background. The visual distraction influences the drivers ability to visualize projected information on the HUD. $\mathbf{E}^{minus}$ represents the degree of reduction of the saliency of the projected information on the HUD. A large reduction of the saliency means that the background hinders the drivers ability to recognize the information on the HUD.

Finally, $\mathbf{E}^{plus}$ and $\mathbf{E}^{minus}$ are converted to scalar values p and m as calculated based on the following

$$p = \frac{\sum_{i=0}^{N-1} \sum_{j=0}^{M-1} E_{i,j}^{plus}}{N \times M \times 255}, \quad (4)$$

$$m = \frac{\sum_{i=0}^{N-1} \sum_{j=0}^{M-1} E_{i,j}^{minus}}{N \times M \times 255}, \quad (5)$$

which are defined as the two evaluation indices. It should be that the saliency in the areas of the projected infor-

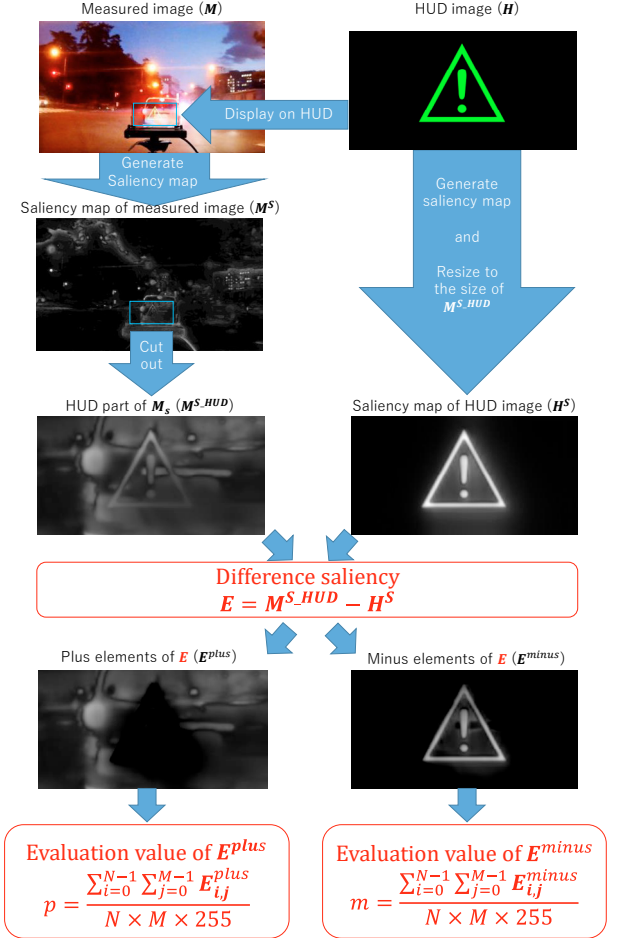


Fig. 1. Difference saliency method

mation on the HUD are different when the information is different. Therefore, the indices can only evaluate the same information with different colors for the same background.

4. EXPERIMENT

The simulation experiment was divided into two parts. The first part considered problems associated with the use of HUDs with different backgrounds, and the second part applied the proposed method to objectively evaluate the mutual interference of saliency between the HUD and background.

4.1 Recording of driving video

Firstly, a video of the front scene was obtained using a camera (GoPro HERO+¹) while a subject was driving a vehicle in an urban area of Nagoya, Japan. The five-hour video included scenes of different time periods such as daytime, evening, and night, in addition to scenes of different traffic environments such as turning at an intersection, driving in a traffic jam, and driving on a crowded road with large pedestrian flow. The video also recorded various glare conditions such as the brake lamps of preceding vehicles, headlights of the oncoming vehicle and sunlight.

¹ GoPro HERO+: <https://gopro.com/update/heroplus>

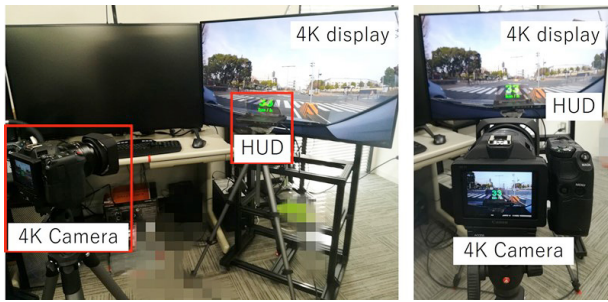


Fig. 2. Experimental equipment.

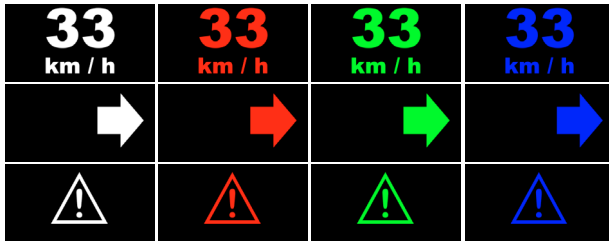


Fig. 3. The prepared information for the HUD.

4.2 Experimental setting

Figure 2 shows the experimental equipment. The still images of the video were replayed on a 42.5-inch 4K display (LG 43UD79T-B²) during the experiment conducted in a laboratory. A 6.2-inch HUD (MAXWIN HUD-622³) was placed in front of the 4K display to simulate the actual driving environment. A 4k camera (Canon XC10⁴) was used to acquire images in the case when overlap occurred between information on the HUD and background. The view angle, elevation angle, and the height of the camera were set close to the actual driving vision of the driver.

Three kinds of information content were projected on the HUD as shown in Fig. 3. Each information had four colors: white, red, green, and blue.

In this experiment, the most basic saliency method proposed by Itti et al. (1998)⁵ was utilized. It should be noted that the motion-based saliency was not considered and the RGB color space was used in this experiment.

4.3 Problems of the HUD used in different backgrounds

There were two typical scenarios that illustrate the mutual interference between the information on the HUD and the background. The first scenario is the case whereby the projected information on the HUD is overly distractive and impacts the drivers ability to observe objects in the surrounding environment. Three examples are shown in Fig. 4. In each example, the left image shows the entire scene from the driver's viewpoint; the upper-right image is the HUD component extracted from the left image; the bottom right image is the HUD component extracted from the saliency map of the left image. These three examples show that pedestrians were blocked either during the



Fig. 4. The projected information on the HUD affected the drivers ability to observe objects in the environment.



Fig. 5. The ambient light attenuated the saliency of the projected information on the HUD.

night-time or the daytime. The HUD component extracted from the saliency map shows that the saliency of the projected information on the HUD was higher than the saliency of the pedestrian component. This situation is likely to increase the risk of an accident when the driver is distracted by the information on the HUD.

The second scenario occurs when the saliency of the projected information on the HUD is reduced in the case of glare. Glare refers to stray light not only from the sun, but also from the brake lights of preceding vehicles and the headlights of oncoming vehicles. Three typical examples are shown in Fig. 5. The HUD components extracted from the saliency map clearly show that the saliency of the projected information on the HUD was significantly lower than the saliency of the background.

² LG 43UD79T-B: <https://www.lg.com/jp/monitor/lg-43UD79T-B>

³ MAXWIN HUD-622: <https://www.maxwin.jp/content/hud/hud-622.html>

⁴ Canon XC10: <https://cweb.canon.jp/prodv/lineup/xc10/>

⁵ An open source of Itti's saliency method that was used in the experiment: <https://github.com/akisato-pySaliencyMap>

4.4 Objective evaluation experiment

In this investigation, the saliency of different color information on the HUD for the same background was objectively evaluated. There were two scenes as shown in the examples in Table 1. One of the scenes was in the daytime and the other was at nighttime. For the HUD images, the same information with different colors had a very similar saliency map although the saliency of blue on a black background was lower than that of the others for human vision. There are two reasons for this: 1) Itti et al. (1998)'s model normalizes feature maps from each color channels. 2) The distances from red, green, blue to the black background color (0, 0, 0) in RGB space is the same.

The objective evaluation results based on the proposed method are shown in Table 1 as the plus values p and the minus values m . The minus values m of the red information were higher than that of the other colors for both of the two scenes. This means that the saliencies of the red information were reduced because the red brake light of the preceding vehicle and the red information of the HUD overlapped with each other. The plus value p represent the visual distraction of information on the HUD. In each scene, the plus values were similar even if the color of the information was different. Compared to the E^{plus} of the two scenes in the Table 1, the white areas show the visual distraction due to the background. The visual distraction in the upper scene was significant compared to the visual distraction of the lower scene because the white area of the upper scene is larger and more pronounced than that of the lower scene. Thus, the plus values p of the upper scene were higher than the plus values p of the lower scene. These results are consistent with the original assumptions in this paper.

5. CONCLUSION

In this paper, a difference saliency method was proposed to objectively evaluate the mutual effect of HUDs for various background. This approach can be used for any saliency map method. A simulation experiment was performed to confirm that the saliency of the information on the HUD is reduced by the light conditions (glare from the sun, the brake light of a vehicle, headlights in the opposite lane, etc.).

Information elements with a red color should be carefully used in HUDs because the saliency of this color is often reduced compared to other colors in some scenes. This occurs because there are many warm light sources in the driving environment, such as brake lights, sunsets, and yellowed streetlights at night. Consequently, the results of this experiment indicate that it is better to avoid assigning red color to emergency information on HUDs in vehicles although, this color is usually for important notifications and warnings in human-machine interfaces.

In the future, the proposed method will be extended to evaluate the saliency of a dynamic scene by extracting the motion feature. Moreover, other color spaces that are closer to human vision, such as LUV and LAB, will be applied in the proposed method.

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Table 1. Objective saliency evaluation of the same information projected on the HUD using different colors for the same background.

Measured image (M)	Saliency map of measured image (M^s)	HUD part of M_s ($M^{s,HUD}$)	HUD image (H)	Saliency map of HUD image (H^s)	Plus elements of $M^{s,HUD} - H^s$ (E^{plus})	Minus elements of $M^{s,HUD} - H^s$ (E^{minus})	Plus value (p)	Minus value (m)
Case 1: an example of using HUD during the daytime								
							0.156	0.009
							0.113	0.067
							0.143	0.016
							0.153	0.019
Case 2: an example of using HUD during the night								
							0.108	0.020
							0.105	0.043
							0.108	0.023
							0.110	0.021